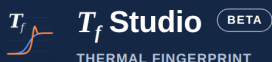




# $T_f$ Studio BETA

THERMAL FINGERPRINT

## User Guide



Every glass carries the **thermal fingerprint** of how it was made, written in its fictive temperature. Tf Studio simulates and fits that history, using the heterogeneous **Takeda-Lucas** model alongside **TNM** and **RelaxPy**, for DSC, MDSC, enthalpy, volume and annealing data.

[Start learning](#)

[Analyze data](#)

[Open Fitter](#)

[Download for Mac / Windows](#)

### TOOLS

#### Learn EXPERIMENTS & MODELS

How structural relaxation is measured, and what the TNM, MAP and Takeda-Lucas models assume, with live simulations throughout.

- Annealing, Kovacs asymmetry and memory, DSC, enthalpy recovery, MDSC ( $C_p^*$ , reversing/non-reversing and their caveats)
- The mathematics of TNM, MAP (RelaxPy) and TL
- Mean field vs Prony vs domain  $T_{f,i}$ : see the distributions

[Open Learn](#) →

#### Data Analysis YOUR DATA

Turn raw DSC heat flow and relaxation data into normalized curves, fictive temperatures and enthalpy.

- Glass and liquid baselines →  $C_p^N$ ,  $T_f(T)$ ,  $T_f$  for heating and cooling scans
- Heating after annealing: aligned scans,  $\Delta H_F$ , recovered  $\Delta H(t_d)$ ,  $\phi$  and a KWW fit
- Volume / density / length during annealing: stretched-exponential fit

[Open Data Analysis](#) →

#### Fitter STEP BY STEP

Program any thermal history, load your measurements, then compute with chosen parameters or fit them.

- Ramps, anneals with log-spaced times, T-jumps, MDSC modulation
- DSC, MDSC  $C_p/C_p^*$ , enthalpy, volume,  $T_f$ , annealing data
- Bounded local / global fits, errors, correlations, AIC/BIC

[Open Fitter](#) →

Documentation: [Quick Start \(PDF, 2 pages\)](#) · [User Guide \(PDF, 25 pages\)](#)

Hands-on: the [Lab](#) runs cool-heat scans, anneals and T-jumps with all three models side by side, with relaxation-time distributions and  $\beta_{KWW}(T)$ .

# Contents

|    |                                                                                                                          |
|----|--------------------------------------------------------------------------------------------------------------------------|
| 1  | What Tf Studio does                                                                                                      |
| 2  | Getting Tf Studio: browser, Mac and Windows, updates                                                                     |
| 3  | Controls shared by every tool                                                                                            |
| 4  | Learn: experiments and models, with live simulations                                                                     |
| 5  | Lab: quick interactive runs                                                                                              |
| 6  | Fitter: thermal history → data → model → compute or fit                                                                  |
|    | 6.1 Thermal history · 6.2 Experimental data · 6.3 Model and parameters · 6.4 Compute and fit · 6.5 Projects and examples |
| 7  | Data Analysis: raw heat flow, enthalpy recovery, property relaxation                                                     |
|    | 7.1 Heat-flow scans · 7.2 Enthalpy recovery · 7.3 Property relaxation · 7.4 From analysis to fit                         |
| 8  | Data files                                                                                                               |
| 9  | Troubleshooting and questions                                                                                            |
| 10 | Reference: models, parameters and data types                                                                             |
| 11 | How to cite, and references                                                                                              |

## In a hurry?

The separate **Quick Start** document takes you from installing to a first result in a few minutes. This guide explains every screen and option in detail.

## 1 · What Tf Studio does

Every glass carries a thermal fingerprint of how it was made, written in its fictive temperature  $T_f$ . Tf Studio simulates that history, analyzes your measurements, and fits relaxation models to them.

Tf Studio contains four tools. They open from the home page (web) or from the launcher (desktop app).

| Tool          | Use it to...                                                                                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Learn         | understand the experiments (annealing, Kovacs jumps, DSC, enthalpy recovery, MDSC) and the models (TNM, MAP / RelaxPy, Takeda–Lucas), with live simulations on every topic. |
| Lab           | try a cool–heat scan, an anneal or a T-jump interactively and compare the three models side by side.                                                                        |
| Fitter        | program any thermal history, load your data, then compute with chosen parameters or fit the parameters, with uncertainties.                                                 |
| Data Analysis | turn raw DSC heat flow into normalized $C_p$ , fictive temperatures and recovered enthalpy, and fit volume (or density, length...) relaxation during annealing.             |

### The three models

| Model                             | In one line                                                                                                                                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Takeda–Lucas (TL)                 | Heterogeneous domains, each with its own Vogel temperature and local fictive temperature $T_{f,i}$ ; the spread of $T_{f,i}$ persists in equilibrium, giving temperature-dependent nonexponentiality and enthalpy fluctuation (J. Chem. Phys. 160, 174504, 2024). |
| Tool–Narayanaswamy–Moynihan (TNM) | One fictive temperature for the whole sample, nonlinearity $x$ and a stretched-exponential (KWW) kernel.                                                                                                                                                          |
| MAP / RelaxPy                     | Mauro–Allan–Potuzak nonequilibrium viscosity with Prony-series components (Wilkinson, Mauro & Mauro, SoftwareX 2018).                                                                                                                                             |

#### Your data stay on your computer

All calculations run locally, in your browser or in the desktop app. Nothing you load is uploaded anywhere.

#### Beta software

Tf Studio is in beta: features, defaults and occasionally results may change between versions. Record the version number (bottom of the launcher) with your results.

## 2 · Getting Tf Studio

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### 2.1 In the browser

Open [wtakeda0129.github.io/tf-studio](https://wtakeda0129.github.io/tf-studio). Nothing to install; it works in Chrome, Edge, Firefox and Safari, and always shows the latest version. After the page has loaded, the tools also work offline.

### 2.2 Desktop app

Download the installer from [github.com/Wtakeda0129/tf-studio/releases/latest](https://github.com/Wtakeda0129/tf-studio/releases/latest), or click **Download for Mac / Windows** on the home page.

#### macOS (Apple silicon and Intel)

- 1 Download `Tf-Studio-<version>-universal.dmg`.
- 2 Open it and drag **Tf Studio** into **Applications** (choose **Replace** if asked).
- 3 Open Tf Studio. The first time, macOS says it cannot verify the developer: click **Done**.
- 4 Open **System Settings** → **Privacy & Security**, scroll down and click **Open Anyway**, then confirm.

This is needed once per computer, because the app is signed with the group's own certificate rather than an Apple Developer ID.

#### Windows 10 / 11

- 1 Download `Tf-Studio-Setup-<version>.exe`.
- 2 Run it. If Windows shows “Windows protected your PC”, click **More info** → **Run anyway**.
- 3 Choose the install folder (no administrator rights are needed) and finish.
- 4 Start Tf Studio from the Start menu or the desktop shortcut.

### 2.3 Updates

The desktop app checks for a new version when it starts and every six hours; you can also check with **Tf Studio** → **Check for Updates...** (Mac) or **Help** → **Check for Updates...** (Windows). A new version downloads in the background and the app offers **Restart now** to install it.

If macOS cannot install an update in place, the app says so and offers **Download**, which opens the release page: install the new DMG as in 2.2. The web version needs no updates.

### 2.4 The launcher

The desktop app opens a launcher with the four tools; each opens in its own window. Shortcuts: `⌘/Ctrl 1` Learn, `2` Lab, `3` Fitter, `4` Data Analysis, `⌘/Ctrl 0` launcher. The version number is shown at the bottom of the launcher.

### 3 · Controls shared by every tool

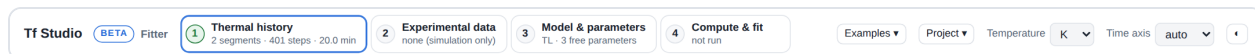


Figure 3.1. Header of the Fitter. The same controls appear in Data Analysis.

| Control                       | What it does                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tf Studio (logo)              | Back to the home page or the launcher.                                                                                                                                                                                                                                                                                          |
| Temperature K / °C            | Unit for every temperature on screen: plots, thermal-history inputs, $T_g$ and its bounds, data limits. Values are stored in kelvin, so switching never changes a calculation.                                                                                                                                                  |
| Time axis auto / linear / log | <i>Auto</i> (default): real (linear) time whenever the temperature changes with time (ramps, MDSC), so scans are shown in real time; log time only for isothermal histories (jump + hold). Relaxation during holds and time-domain hold data stay on log time. <i>Linear</i> or <i>log</i> force one scale for every time axis. |
| ☐                             | Switches between the light (default) and dark colour scheme.                                                                                                                                                                                                                                                                    |
| SVG (on each plot)            | Saves that plot as a vector graphic for slides or papers.                                                                                                                                                                                                                                                                       |
| Hover on a plot               | Shows the nearest data point and its values.                                                                                                                                                                                                                                                                                    |

#### Typing numbers

Inputs accept any number format ( `1e5` , `0.5` , `-40` ). Temperatures are read in the unit selected in the header.

## 4 · Learn

Learn is a self-contained course on structural relaxation. The contents on the left jump to any section; **Part I** covers experiments and **Part II** the models.

Tf Studio BETA Learn

TL TNM MAP Experiments Models Lab Fitter

CONTENTS

1 · Fictive temperature

PART I · EXPERIMENTS

2 · Isothermal annealing

3 · Kovacs: asymmetry, memory

4 · DSC cooling and heating

5 · Enthalpy recovery

6 · MDSC and  $C_p^*$

PART II · MODELS

7 · TNM

8 · MAP / RelaxPy

9 · Takeda–Lucas

10 · See the difference

11 · Side by side

12 · Further reading

### Learn: glass relaxation, from experiment to model

Part I covers the experiments used to study structural relaxation: annealing, Kovacs jumps, DSC, enthalpy recovery and modulated DSC. Each one comes with a live simulation. Part II explains the mathematics of the three models and where they differ. TNM gives the whole sample one fictive temperature. MAP splits the relaxation into Prony components, and all of them return to the same temperature in equilibrium. The Takeda–Lucas model gives every domain its own fictive temperature, so the fictive temperatures stay spread out even in the equilibrium liquid.

#### 1 Fictive temperature and structural relaxation

Above the glass transition, a liquid's structure keeps pace with temperature. On cooling, the structural relaxation time  $\tau$  grows so quickly that the structure eventually cannot keep up and freezes. Tool described the frozen structure by a **fictive temperature**  $T_f$ : the temperature at which the liquid would have this structure in equilibrium. In equilibrium  $T_f = T$ . In the glass,  $T_f$  records the thermal history, which is why Tf Studio calls it the *thermal fingerprint*.

Enthalpy and volume follow  $T_f$ . With liquid and glass heat capacities  $C_{p,l}$  and  $C_{p,g}$ , the normalized heat capacity measured by DSC is

$$C_p^N(T) = \frac{C_p - C_{p,g}}{C_{p,l} - C_{p,g}} = \frac{dT_f}{dT}.$$

Relaxation towards equilibrium has three well-known signatures, and every model below has to reproduce all three:

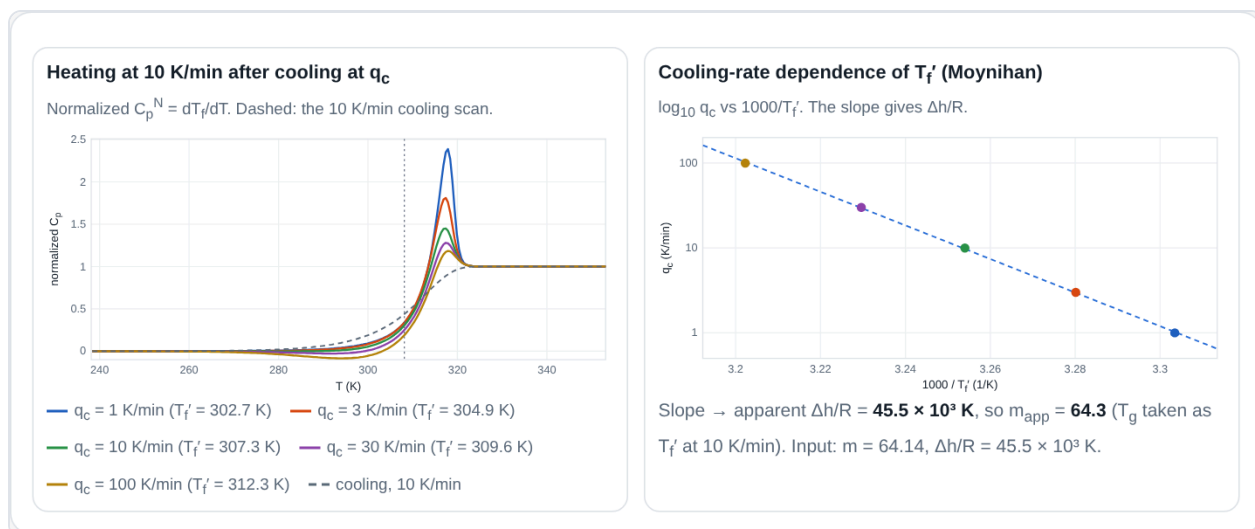
- **Nonexponential.** After a small temperature step, the response follows a stretched exponential (Kohlrausch–Williams–Watts, KWW),  $\Psi(t) = \exp[-(t/\tau)^\beta]$  with  $0 < \beta \leq 1$ .
- **Nonlinear.**  $\tau$  depends on the structure ( $T_f$ ) as well as on  $T$ . A glass therefore relaxes faster when it approaches equilibrium from above than from below (Kovacs asymmetry).
- **Non-Arrhenius.** In equilibrium,  $\tau(T)$  follows a VFT, Adam–Gibbs or MYEGA form, and its steepness at  $T_g$  is the fragility  $m = d \log_{10} \tau / d(T_g - T)$ .

**Figure 4.1.** Learn. The switch at the top (TL / TNM / MAP) chooses the model used by all simulations in Part I.

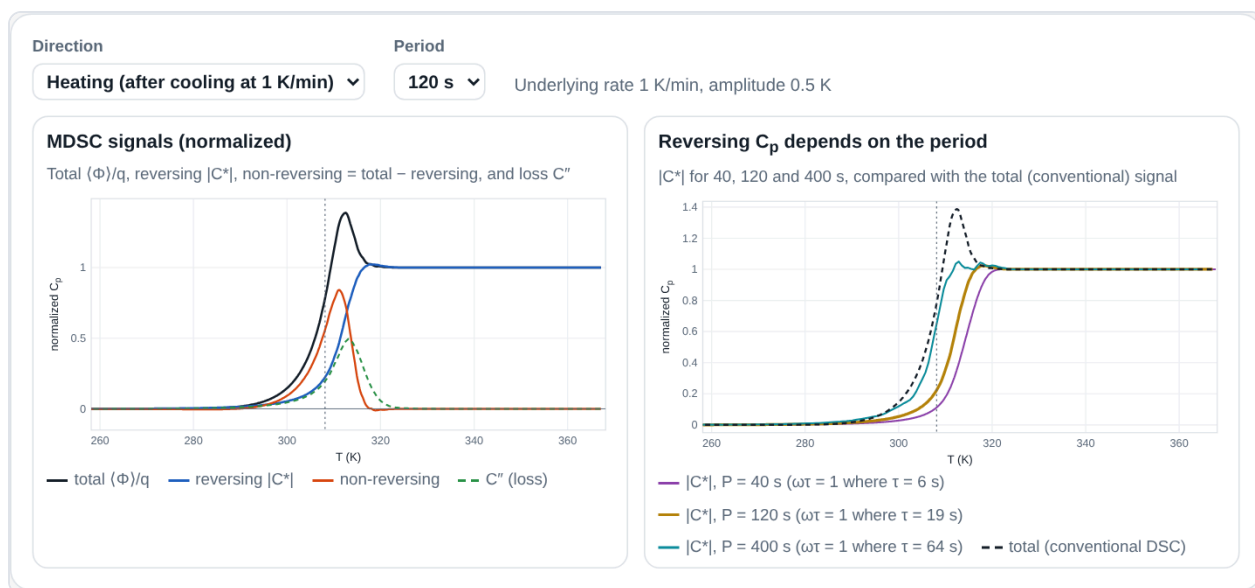
### Part I · Experiments

Each experiment has its protocol and equations, then a live simulation of the same example glass, selenium ( $T_g = 308.13$  K,  $m = 64.14$ , with the TL parameters of Takeda & Lucas 2024; for MAP the glassy viscosity is Arrhenius with  $B = x m T_g$ ,  $x = 0.5$ , and  $C = 0$ ):

- **Isothermal annealing:** relaxation function  $\phi(t)$  after a 10 K down-jump, compared with a linear KWW curve, and the apparent KWW parameters.
- **Kovacs experiments:** asymmetry of approach ( $\pm 5$  K jumps) and the memory effect (Kovacs; Macedo–Napolitano crossover).
- **DSC on cooling and heating:** heating scans after five cooling rates and Moynihan's cooling-rate plot (apparent  $\Delta h/R$  and fragility).
- **Enthalpy recovery:** heating scans after annealing and  $\Delta H(t_a)$ .
- **MDSC:** total, reversing and non-reversing signals,  $C''$ , and how the reversing step moves with the modulation period (the Hutchinson and Schawe caveats).



**Figure 4.2.** DSC section: heating after five cooling rates (left) and the cooling-rate dependence of  $T_f'$  (right).



**Figure 4.3.** MDSC section. Choose **Direction** (heating or cooling) and **Period**; the right plot compares the reversing  $C_p$  for three periods.

## Part II • Models

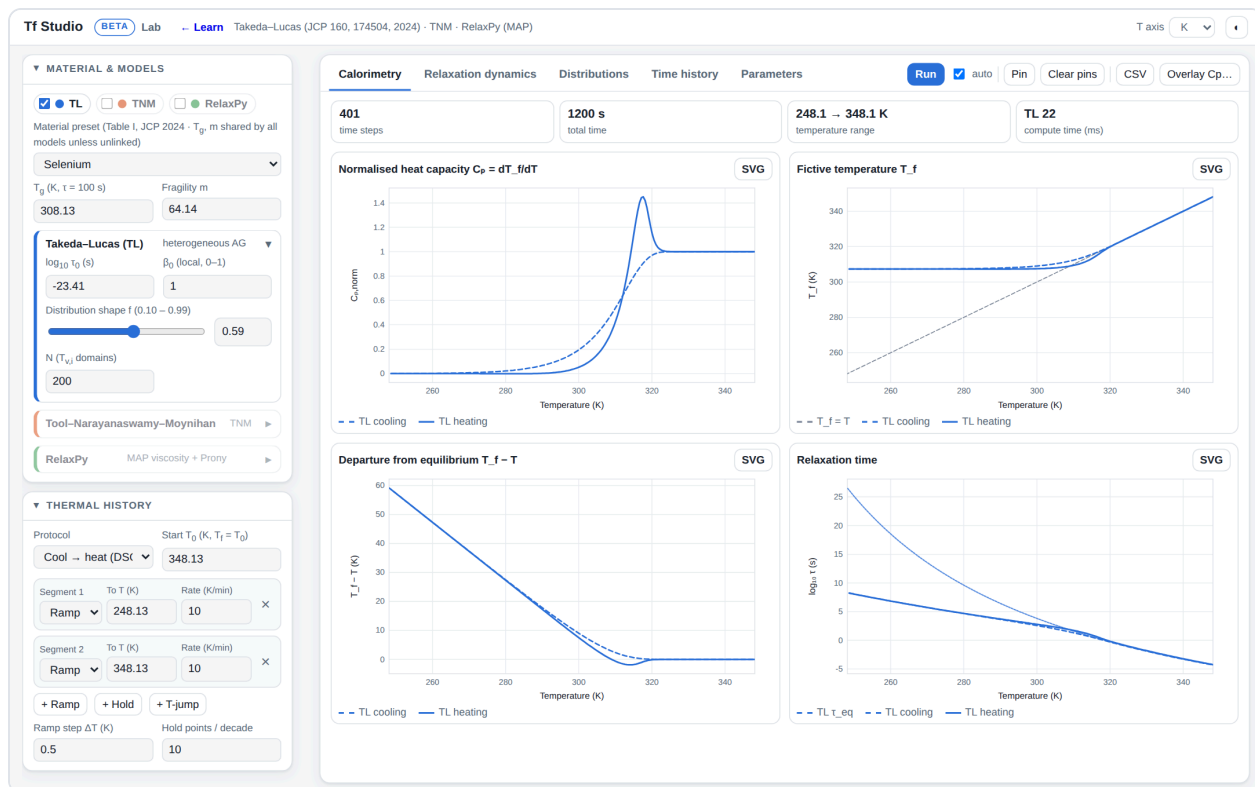
The mathematics of TNM, MAP and TL, what each assumes, and an interactive comparison of where the fictive temperatures sit.



**Figure 4.4.** “See the difference”: the three models run the same cool–anneal history. Drag the slider or press ▶ **Play**; change the cooling rate, annealing temperature and TL width  $f$ .

## 5 · Lab

The Lab (formerly Explorer) is a quick, always-live playground: change a number and the plots update. Use it to build intuition; use the Fitter for your own histories and data.



**Figure 5.1.** Lab: models and material on the left, protocol below; tabs on the right.

- **Models:** tick TL, TNM and/or RelaxPy. **Material preset** loads the TL parameters of Table I of the JCP 2024 paper;  $T_g$  and  $m$  are shared by the models unless unlinked in their sections.
- **Thermal history:** choose a protocol (cool → heat, slow cool, anneal, Kovacs T-jump) or build segments with **+ Ramp**, **+ Hold**, **+ T-jump**.
- **Tabs:** Calorimetry ( $C_p$ ,  $T_f$ ), Relaxation dynamics, Distributions (TL relaxation-time distributions and  $\beta_{KWW}(T)$ ), Time history, Parameters.
- **Run / auto:** untick **auto** to update only when you click **Run**. **Pin** keeps a run for comparison; **CSV** exports the curves.

# 6 · Fitter

The Fitter works in four steps, shown as buttons across the top. You can move between them at any time.

| Step                 | What you do                                                    | Right panel shows                |
|----------------------|----------------------------------------------------------------|----------------------------------|
| 1 Thermal history    | Build the temperature program.                                 | T(t) and the time steps          |
| 2 Experimental data  | Optional: add measurements.                                    | your data and the linked segment |
| 3 Model & parameters | Choose a model, values, which parameters are free, and bounds. | history and data                 |
| 4 Compute & fit      | Click <b>Compute</b> or <b>Run fit</b> .                       | all results                      |

Nothing is calculated until you click

Steps 1–3 only show the thermal history and your data. The model runs when you click **Compute** or **Run fit** in step 4. Any later change clears the old result and asks you to compute again, so a result on screen always matches the current settings.

## 6.1 Step 1 · Thermal history

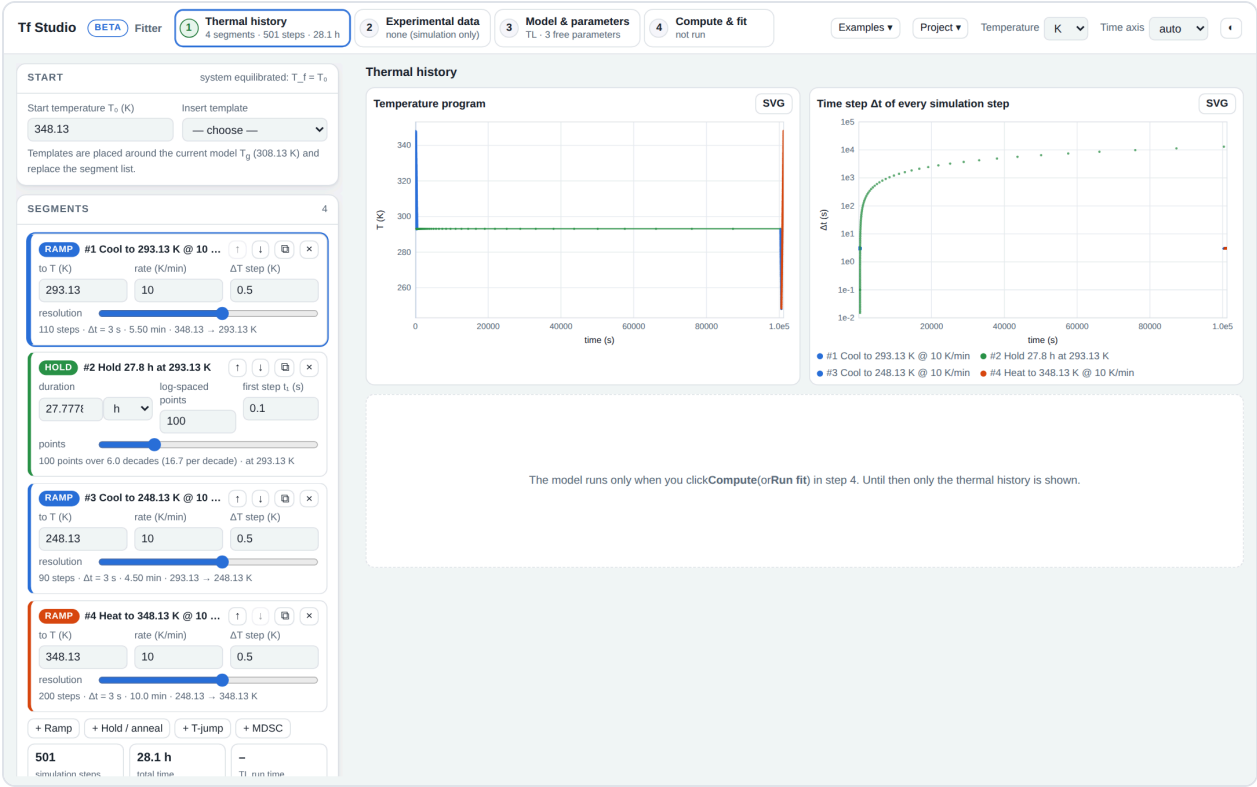


Figure 6.1. Step 1 with the template “Aging below  $T_g$ , then DSC heating”.

- 1 **Start temperature  $T_0$ .** The sample starts in equilibrium at  $T_0$  ( $T_f = T_0$ ). Choose  $T_0$  above  $T_g$ .
- 2 **Insert template** (optional). Templates are placed around the current model  $T_g$ , so set  $T_g$  in step 3 first if your material differs from the default.
- 3 **Add segments** with **+ Ramp**, **+ Hold / anneal**, **+ T-jump**, **+ MDSC**. Reorder, duplicate or delete them with the small buttons on each card.

| Segment       | Inputs                                                                   | Notes                                                                            |
|---------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Ramp          | target T, rate (K/min), $\Delta T$ step                                  | 0.1–0.5 K steps are accurate for DSC; the slider sets $\Delta T$ on a log scale. |
| Hold / anneal | duration (s, min, h, d), log-spaced points, first step $t_1$             | Points are spaced logarithmically from $t_1$ to the duration.                    |
| T-jump        | target T                                                                 | Instantaneous change (e.g. Kovacs experiments).                                  |
| MDSC          | target T, underlying rate, amplitude $\pm A$ , period, points per period | Rate 0 gives quasi-isothermal MDSC with a duration.                              |

Under the segments, the summary shows the number of simulation steps and the total time. The right panel plots the temperature program and the time step of every simulation step.

## 6.2 Step 2 · Experimental data (optional)

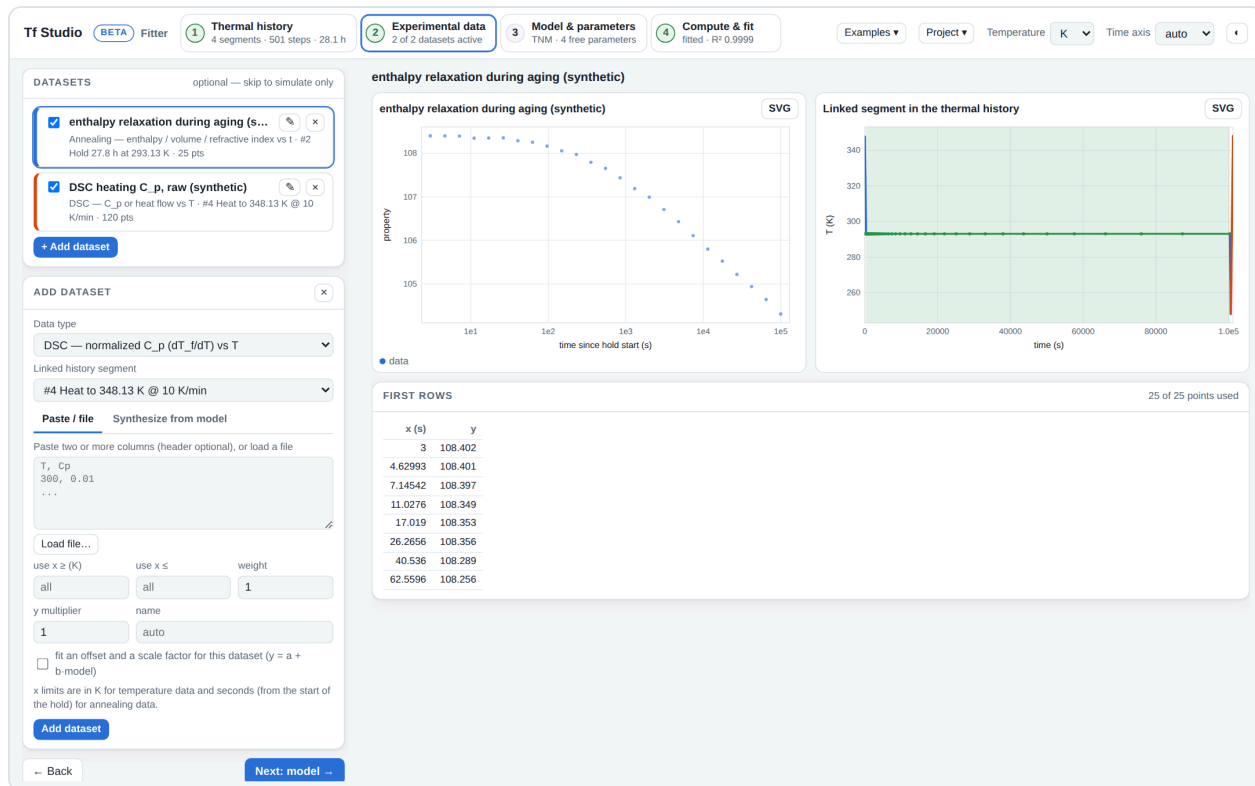


Figure 6.2. Step 2 with two datasets and the **Add dataset** form open.

- 1 Click **+ Add dataset** and choose the **Data type** (table in section 10).
- 2 Choose the **Linked history segment**: the part of the thermal history the data belong to (e.g. the heating ramp for a DSC heating scan, the hold for annealing data).
- 3 **Paste / file**: paste two or more columns or click **Load file...**, then pick the x and y columns and units. Or **Synthesize from model** to generate test data with noise.
- 4 Optional: **use  $x \geq$  /  $\leq$**  to limit the fitted range, a **weight**, a **y multiplier**, and **fit an offset and a scale factor** ( $y = a + b \cdot \text{model}$ ) when the data are not normalized.
- 5 Click **Add dataset**. Tick or untick a dataset to include it; edits it, removes it.

### Raw data are fine

Types such as “C<sub>p</sub> or heat flow vs T”, “Enthalpy vs T” and “property vs t” fit their baselines or linear coefficients exactly at every step, so you can load raw measurements. To normalize DSC data yourself first, use Data Analysis (section 7).

## 6.3 Step 3 · Model and parameters

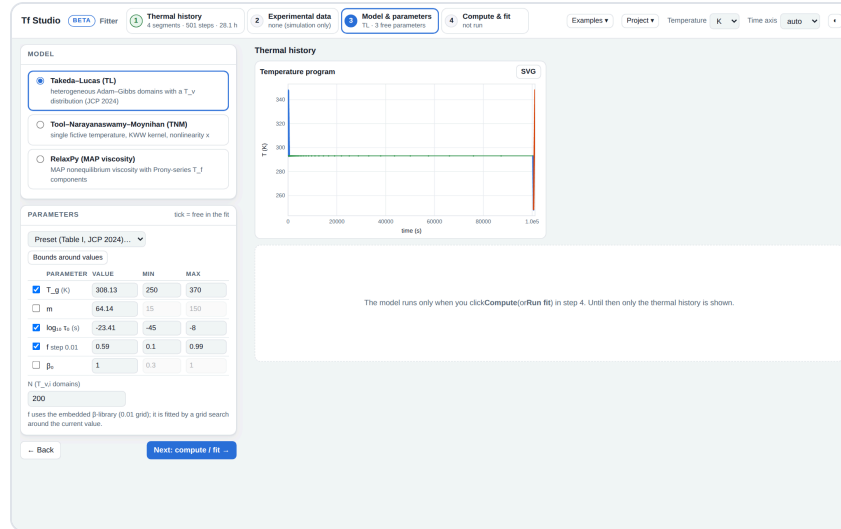


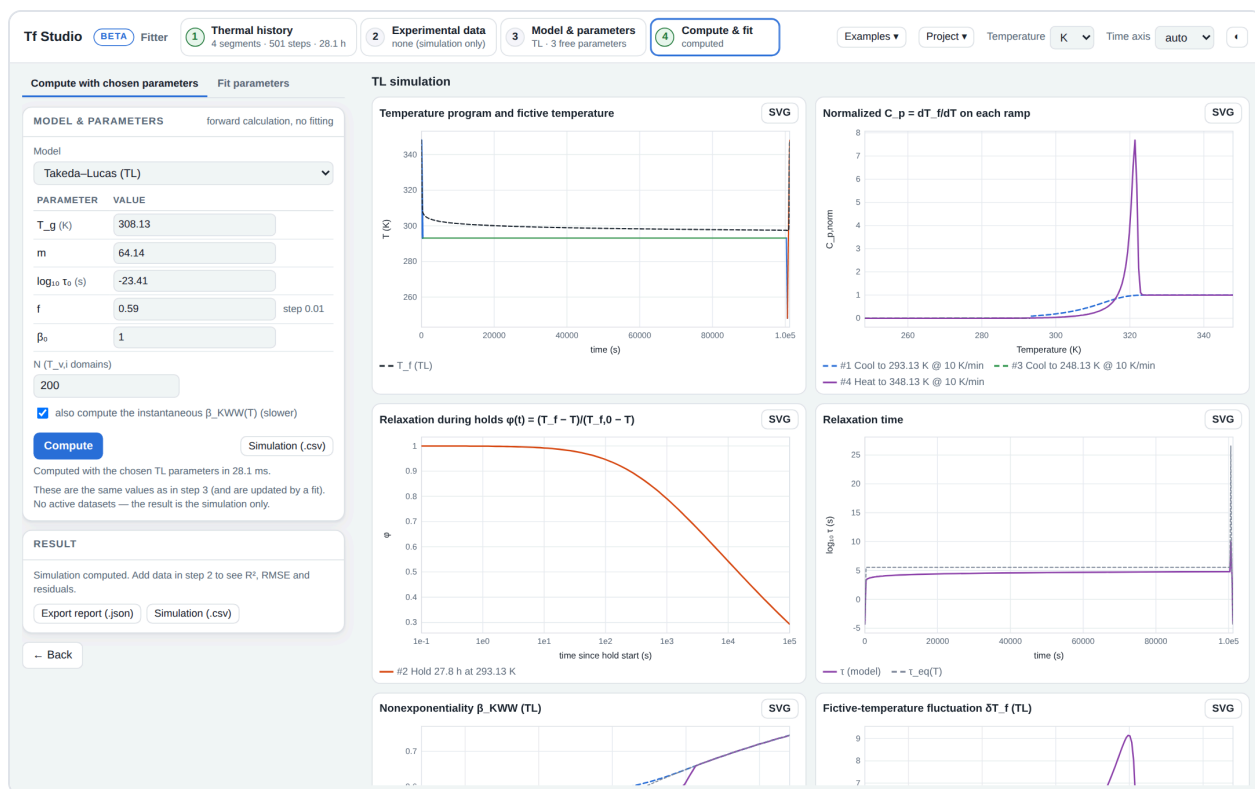
Figure 6.3. Step 3: model, parameter values, tick = free in the fit, and fit bounds.

- Choose **Takeda–Lucas**, **TNM** or **RelaxPy (MAP)**.
- Type values. Ticked parameters are **free** in a fit and move between **min** and **max**; unticked ones stay fixed. **Bounds around values** sets sensible bounds.
- TL: **Preset** loads Table I of the JCP 2024 paper; **N** is the number of domains (200 is accurate). *f* is fitted on its 0.01 grid.
- TNM: choose the exact KWW summation or the Prony kernel for display (fits always use Prony).
- MAP: **RelaxPy example glass**; *A* can follow from continuity at  $T_g$ , and  $p = 0.3082 \cdot m$  (RelaxPy default).

## 6.4 Step 4 · Compute and fit

### Compute with chosen parameters

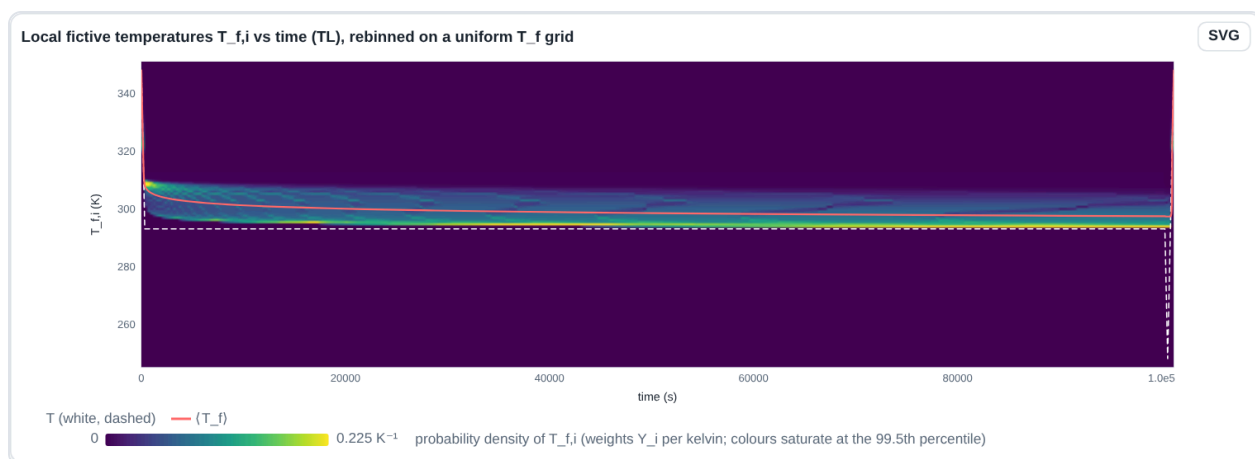
Forward calculation, no fitting. Check or adjust the values and click **Compute**. With data, the agreement ( $R^2$ , RMSE per dataset) is shown.



**Figure 6.4.** Result of **Compute**: temperature program and  $T_f$ , normalized  $C_p$  on each ramp, relaxation during holds, and relaxation time.

### Extra results for the TL model

- **Fictive-temperature fluctuation  $\delta T_f$  vs temperature**, with the equilibrium value.
- **Map of the local fictive temperatures  $T_{f,i}$  vs time**, rebinned on a uniform  $T_f$  grid (probability density in  $K^{-1}$ ), with  $T(t)$  and  $\langle T_f \rangle(t)$ .
- **β<sub>KWW</sub>(T)**: tick **also compute the instantaneous β<sub>KWW</sub>(T)** (adds about a second).



**Figure 6.5.** TL: local fictive temperatures during cooling, aging and reheating.



Figure 6.6. TL: instantaneous  $\beta_{KWW}$  along each segment and in equilibrium (grey).

## Fit parameters



Figure 6.7. Fit result: data with the fitted model and residuals, convergence, parameter correlation, and the parameter table with standard errors.

- 1 Open the **Fit parameters** tab (needs at least one active dataset and one free parameter).
- 2 Choose the **Method**: *Local — Levenberg–Marquardt* (fast, starts from the current values) or *Global — differential evolution + LM polish* (slower, searches the whole box).
- 3 Set **Max model runs** and the **Weighting** (each dataset equal, or each point equal).
- 4 Click **Run fit**. **Stop** ends early; **Undo fit** restores the previous values.

The result lists each parameter with  $\pm 1$  standard error, flags parameters *at bound* or *poorly determined*, and gives  $R^2$ , reduced  $\chi^2$ , AIC and BIC. In the correlation matrix,  $|r| > 0.95$  (bold) means two parameters trade off and are not separately determined by your data.

## Exports

**Export report (.json)** (parameters, errors, statistics, history), **Model vs data (.csv)**, **Simulation (.csv)** (t, T,  $T_f$ ,  $\tau$ ,  $\tau_{eq}$ , segment), and the **SVG** button on each plot.

## 6.5 Projects and examples

**Project ▼**: *New project*, *Save project (.json)* (history, data, model, values, bounds, fit options) and *Open project....* **Examples ▼** loads ready-made cases: a selenium DSC cycle, aging + DSC with synthetic enthalpy data, and MDSC with synthetic  $C_p'/C_p''$ .

## 7 · Data Analysis

Data Analysis prepares your own measurements. It has three tabs across the top. Each tab has **Load example data** (synthetic selenium data made with the TL model, parameters of Se from Takeda & Lucas 2024) so you can try everything first.

### 7.1 Heat-flow scans

#### ADD A HEAT-FLOW RUN

File (.csv, .txt, instrument export)

Choose File No file chosen

...or paste columns

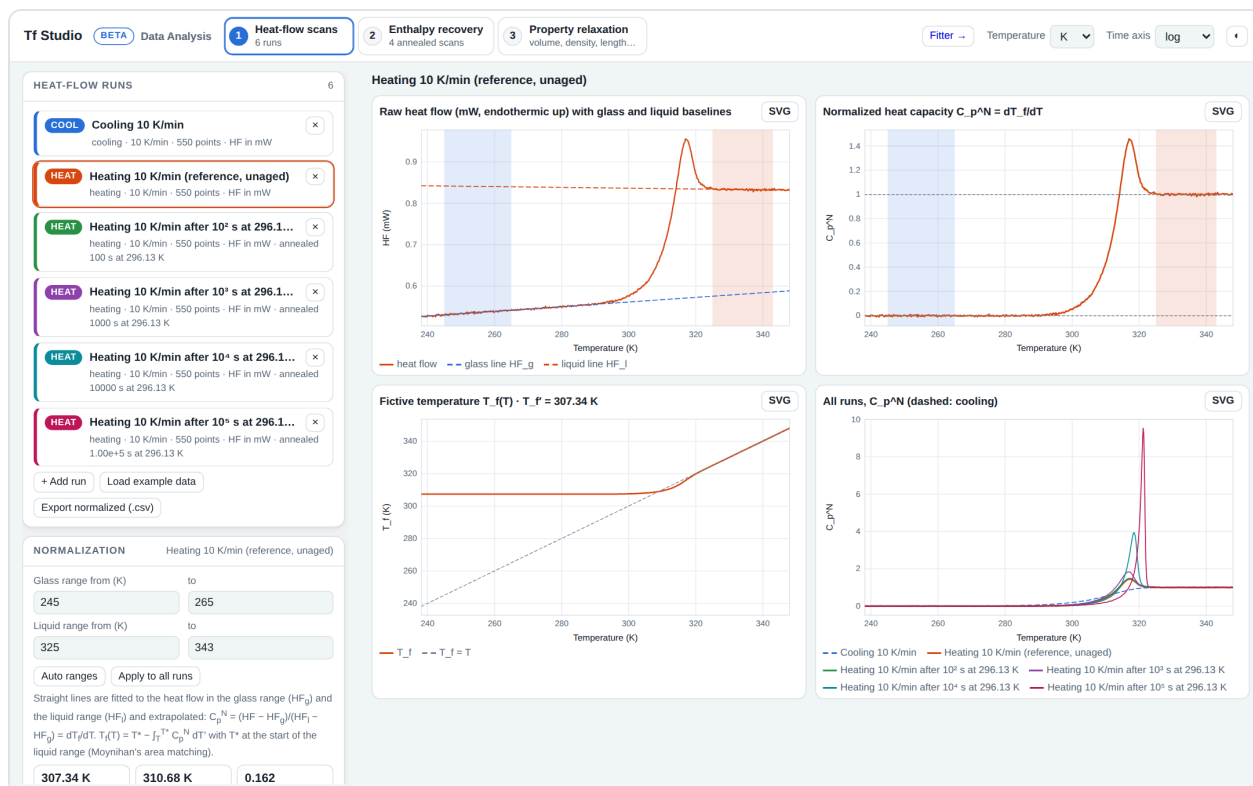
```
Time (min),Temperature (°C),Heat Flow (W/g)
0.00,-120.0,-0.1000
0.05,-119.5,-0.1000
0.10,-119.0,-0.1000
0.15,-118.5,-0.1000
...
```

161 numeric rows, 3 columns detected.

|                                    |                              |
|------------------------------------|------------------------------|
| Temperature column                 | Heat-flow column             |
| <div>Temperature (°C) ▼</div>      | <div>Heat Flow (W/g) ▼</div> |
| Temperature unit                   | Heat-flow unit               |
| <div>°C ▼</div>                    | <div>W/g ▼</div>             |
| Sign convention                    | Scan                         |
| <div>endothermic down (exc ▼</div> | <div>auto (heating) ▼</div>  |
| Rate  q  (K/min)                   | Annealing time $t_a$ (s)     |
| <div>10</div>                      | <div>none</div>              |
| Annealing $T_a$ (K)                |                              |
| <div>—</div>                       |                              |
| Name                               |                              |
| <div>auto</div>                    |                              |
| <div>Add run</div>                 | <div>Cancel</div>            |

**Figure 7.1.** Adding a run: paste or load a file, then check the detected columns and units.

- 1 Click **+ Add run**, then choose a file (.csv, .txt, instrument export) or paste columns.
- 2 Check the **Temperature** and **Heat-flow** columns and their units (K or °C; mW, W/g or arbitrary). Column names like “Temperature (°C)” and “Heat Flow (W/g)” set these automatically.
- 3 Set the **Sign convention**: *endothermic up*, or *endothermic down (exo up)* as in TA Instruments exports.
- 4 **Scan** is detected from the temperature trend (heating/cooling). Enter the **rate** and, for mW, the **sample mass**.
- 5 For a heating scan measured after annealing, enter the **annealing time**  $t_a$  and  $T_a$ . Leave them empty for the unaged reference.
- 6 Click **Add run**.



**Figure 7.2.** Normalization of the reference heating scan. Blue band: glass range; orange band: liquid range; dashed: the fitted baselines.

**Normalization.** Enter a **glass range** and a **liquid range** (or **Auto ranges**). A straight line is fitted to the heat flow in each range and extrapolated, giving

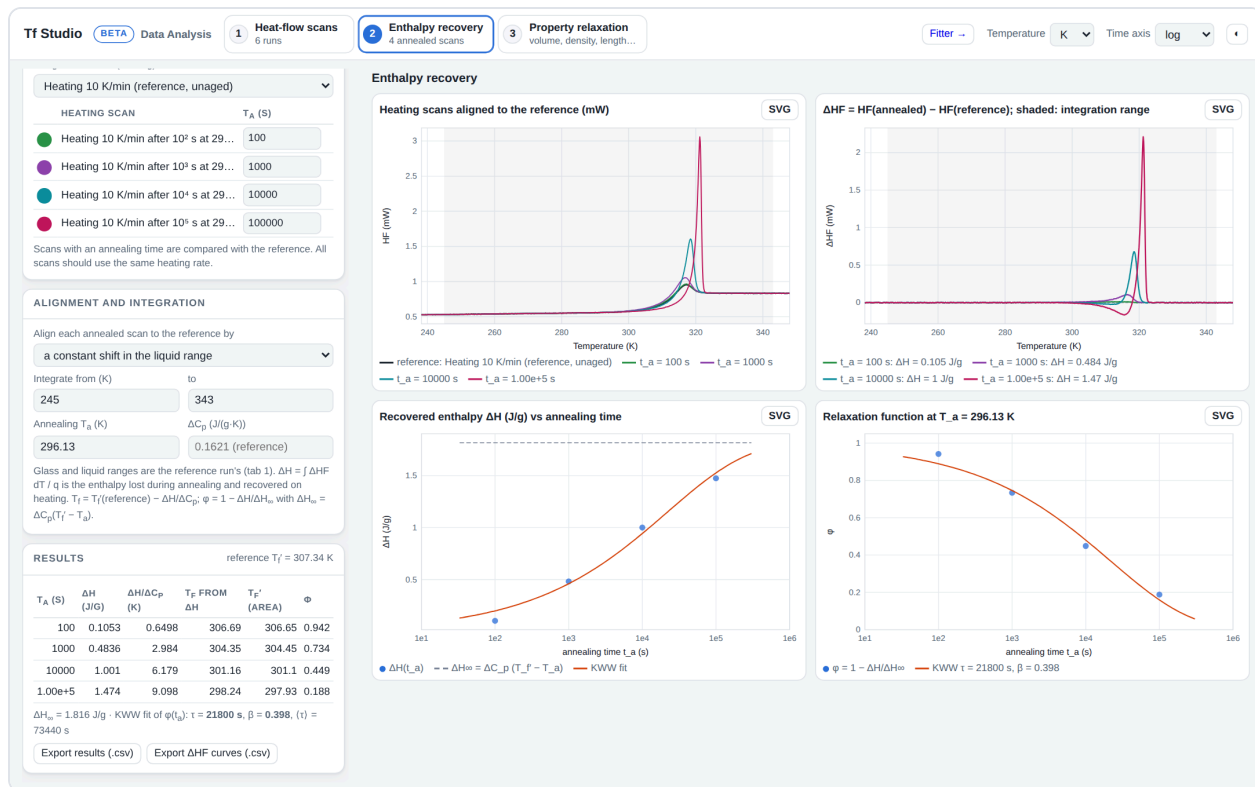
$$C_p^N = (HF - HF_g) / (HF_l - HF_g) = dT_f/dT, \quad T_f(T) = T^* - \int_T^{T^*} C_p^N dT'$$

with  $T^*$  at the start of the liquid range (Moynihan's area matching). The card reports the limiting fictive temperature  $T_f'$ , the **midpoint** ( $C_p^N = 0.5$ ) and  $\Delta C_p$  at  $T_f'$ . **Apply to all runs** copies the ranges; **Export normalized (.csv)** saves  $T$ ,  $HF$ ,  $C_p^N$ ,  $T_f$  for every run.

## Choosing the ranges

- Glass range: well below the transition, where the scan is straight; avoid the start-up transient.
- Liquid range: above the overshoot, where the scan is straight again.
- For comparisons (cooling rates, aging series), use the same ranges for all runs.

## 7.2 Enthalpy recovery



**Figure 7.3.** Enthalpy recovery: aligned scans, ΔHF(T), recovered ΔH vs annealing time, and φ(t<sub>a</sub>) with a KWW fit.

- 1 Choose the **Unaged reference** (a heating scan without annealing).
- 2 Enter or check the annealing time **t<sub>a</sub>** of each annealed heating scan.
- 3 Choose how each annealed scan is **aligned** to the reference: a constant shift in the liquid range, or a straight line through the glass and liquid ranges (use this if the glass lines differ slightly).
- 4 Check the **integration range** (default: from the glass range to the liquid range) and enter **T<sub>a</sub>**.
- 5 Optionally override **ΔC<sub>p</sub>** (default: the reference at T<sub>f</sub>').

$$\Delta H = (1/q) \int \Delta HF \, dT, \quad T_f = T_f'(\text{ref}) - \Delta H / \Delta C_p, \quad \phi = 1 - \Delta H / \Delta H_{\infty}, \quad \Delta H_{\infty} = \Delta C_p(T_f' - T_a)$$

| Result column                          | Meaning                                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta H$                             | enthalpy lost during annealing and recovered on heating (J/g with mW + mass or W/g; mJ with mW and no mass)                        |
| $\Delta H/\Delta C_p$                  | the same in kelvin: the drop of the fictive temperature during annealing                                                           |
| $T_f$ from $\Delta H / T_f'$<br>(area) | fictive temperature after annealing, from $\Delta H$ and independently from area matching of the annealed scan (they should agree) |
| $\phi$                                 | relaxation function; with three or more annealing times a KWW fit gives $\tau$ , $\beta$ and $\langle \tau \rangle$                |

### Use the same heating rate

$\Delta H$  is computed with the reference's rate. All scans should be heated at the same rate and ideally measured on the same sample.

## 7.3 Property relaxation

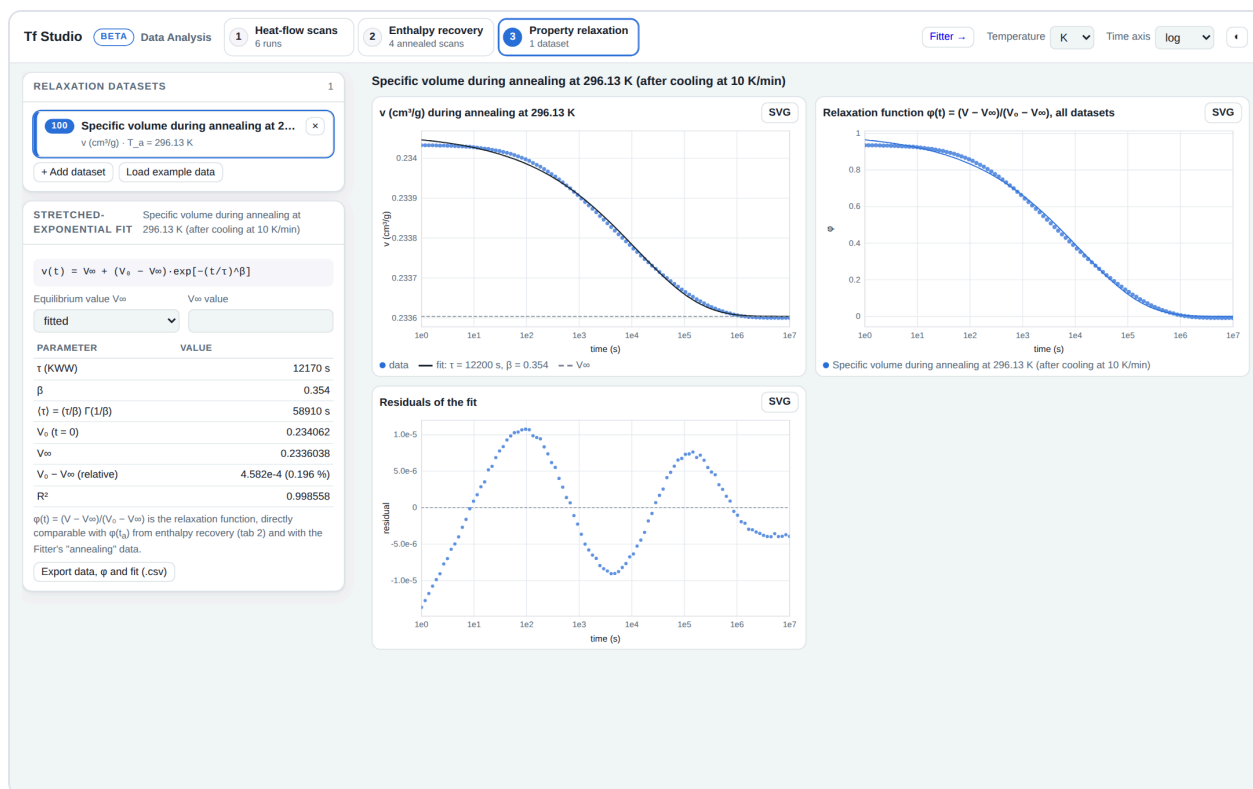


Figure 7.4. Specific volume during annealing with the stretched-exponential fit,  $\phi(t)$  and residuals.

Add a dataset of a property against time during isothermal annealing: volume, density, length, refractive index... (+ **Add dataset**: pick the time column and unit, the property column, a label and  $T_a$ ). The fit is

$$V(t) = V_{\infty} + (V_0 - V_{\infty}) \exp[-(t/\tau)^{\beta}]$$

with  $V_\infty$  **fitted** or **fixed** at a value you enter (use fixed when the data stop well before equilibrium). The table gives  $\tau$ ,  $\beta$ ,  $\langle \tau \rangle = (\tau/\beta)\Gamma(1/\beta)$ ,  $V_0$ ,  $V_\infty$ , the relative change and  $R^2$ . Several datasets can be compared on the  $\phi(t)$  plot.

Systematic residuals (as in Figure 7.4) are expected for real glasses: relaxation is nonlinear, so a single stretched exponential is only an approximation. Fitting the model in the Fitter accounts for this.

### 7.4 From analysis to fit

Export the normalized curves (tab 1) or  $\phi(t)$  (tabs 2, 3) as CSV and paste them into Fitter step 2 as “DSC — normalized  $C_p$ ” or “Annealing — normalized relaxation  $\phi(t)$ ”, linked to the matching segment of a thermal history that reproduces your experiment.

## 8 · Data files

- **Formats:** text files (.csv, .txt, .dat, .tsv) or pasted text. Columns may be separated by commas, semicolons, tabs or spaces.
- **Headers:** lines before the numbers (titles, column names, unit rows) are skipped; the last text line before the data is used for column names.
- **Decimal point:** use a period ( 0.25 ). Files with decimal commas must be converted first.
- **Order:** any; cooling scans may run from high to low temperature.
- **Instrument exports:** export to text/CSV from the instrument software (e.g. TA Universal Analysis / TRIOS, Netzsch Proteus, Mettler STARE) with temperature and heat flow columns.

| Example (DSC, TA style)                     | Example (annealing) |
|---------------------------------------------|---------------------|
| Time (min),Temperature (°C),Heat Flow (W/g) | t (s) V (cm3/g)     |
| 0.00, -120.0, -0.1000                       | 1 0.795716          |
| 0.05, -119.5, -0.1001                       | 1.2 0.795715        |
| ...                                         | ...                 |

## 9 · Troubleshooting and questions

| Problem                                                          | What to do                                                                                                                                   |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| The Fitter shows no model curves.                                | Expected: the model runs only when you click <b>Compute</b> or <b>Run fit</b> in step 4.                                                     |
| A plot shows “Could not draw ...”.                               | The rest of the page still works. Check the inputs of that part (e.g. a range with fewer than two points) and report it if it persists.      |
| A hold shows no relaxation.                                      | The hold is above $T_g$ : the liquid equilibrates within milliseconds. Put the hold below $T_g$ .                                            |
| Mac: “cannot be opened / cannot verify the developer”.           | System Settings → Privacy & Security → <b>Open Anyway</b> (once per computer).                                                               |
| Mac: “Restart now” does nothing, or the version does not change. | Download the latest DMG and replace the app (section 2.2). Versions before 1.1.6 cannot update themselves.                                   |
| Windows: “Windows protected your PC”.                            | <b>More info</b> → <b>Run anyway</b> (first install only).                                                                                   |
| TL map or $\beta_{KWW}$ plot missing.                            | Very long histories (above ~60 000 steps with 200 domains) do not keep every $T_{f,i}$ . Use a coarser $\Delta T$ step or fewer hold points. |
| A fitted parameter is “at bound”.                                | Widen its bounds in step 3, or fix it if the data cannot determine it.                                                                       |
| Parameters strongly correlated ( $ r  > 0.95$ ).                 | Add data that separate them (several cooling rates, annealing data) or fix one of them.                                                      |
| $C_p^N$ is upside down.                                          | Change the <b>sign convention</b> of the run (endothermic up/down).                                                                          |
| $\Delta H$ has the wrong sign or size.                           | Check the sign convention, the rate and the sample mass, and that all scans used the same heating rate.                                      |

Report problems or ideas at [github.com/Wtakeda0129/tf-studio/issues](https://github.com/Wtakeda0129/tf-studio/issues) or to the Lucas group.

## 10 · Reference

### Model parameters

| Model         | Parameter               | Meaning                                                                                        |
|---------------|-------------------------|------------------------------------------------------------------------------------------------|
| TL            | $T_g$                   | temperature where $\langle \tau \rangle = 100$ s                                               |
|               | $m$                     | fragility index                                                                                |
|               | $\log_{10} \tau_0$      | high-temperature limit of $\tau$ (s)                                                           |
|               | $f$                     | width of the relaxation-time distribution ( $\approx \beta_{\text{KWW}}$ at $T_g$ ), 0.10–0.99 |
|               | $\beta_0$               | intrinsic (local) stretching; 1 = fully heterogeneous                                          |
|               | $N$                     | number of domains (numerical; 200 is accurate)                                                 |
| TNM           | $T_g$                   | $\tau(T_g) = 100$ s                                                                            |
|               | $m$                     | fragility; $\Delta h/R = m \ln 10 T_g$                                                         |
|               | $x$                     | nonlinearity (0–1)                                                                             |
|               | $\beta$                 | KWW stretching exponent                                                                        |
| MAP / RelaxPy | $T_g$                   | $\eta(T_g) = 10^{12}$ Pa·s                                                                     |
|               | $m$                     | fragility (MYEGA)                                                                              |
|               | $\log_{10} \eta_\infty$ | high-temperature viscosity limit                                                               |
|               | $A, B, C$               | nonequilibrium viscosity branch; $A$ from continuity at $T_g$ by default                       |
|               | $p$                     | ergodicity exponent (default $0.3082 \cdot m$ )                                                |
|               | $\log_{10} K_s$         | modulus in $\tau = \eta/K_s$                                                                   |
|               | $\beta$                 | stretching of the Prony series                                                                 |

## Fitter data types

| Data type                                             | Linked to  | Notes                                        |
|-------------------------------------------------------|------------|----------------------------------------------|
| DSC — normalized $C_p$ ( $dT_f/dT$ ) vs T             | ramp       | already normalized (e.g. from Data Analysis) |
| DSC — $C_p$ or heat flow vs T                         | ramp       | raw; glass/liquid baselines fitted           |
| Enthalpy vs T                                         | ramp, MDSC | $H = a + bT + \Delta C_p T_f$ fitted         |
| Volume / density / length vs T                        | ramp, MDSC | dilatometry                                  |
| Fictive temperature $T_f$ vs T                        | ramp, MDSC |                                              |
| Annealing — normalized $\varphi(t)$                   | hold       | $\varphi = (T_f - T)/(T_{f,0} - T)$          |
| Annealing — $T_f(t)$                                  | hold       |                                              |
| Annealing — enthalpy / volume / refractive index vs t | hold       | $a + b \cdot T_f$ fitted                     |
| MDSC — $C_p'$ (reversing), $C_p''$                    | MDSC       | normalized; optional scale                   |

## Glossary

|              |                                                                                           |
|--------------|-------------------------------------------------------------------------------------------|
| $T_f$        | fictive temperature: the temperature at which the liquid would have the current structure |
| $T_f'$       | limiting fictive temperature of the glass (frozen value after cooling)                    |
| $C_p^N$      | normalized heat capacity, 0 in the glass and 1 in the liquid; equals $dT_f/dT$            |
| $\varphi(t)$ | normalized relaxation function, $1 \rightarrow 0$ during annealing                        |
| $KWW$        | stretched exponential $\exp[-(t/\tau)^\beta]$                                             |
| $\delta T_f$ | spread (standard deviation) of the local fictive temperatures in the TL model             |

## 11 · How to cite, and references

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If Tf Studio contributes to your work, please cite:

- W. Takeda and P. Lucas, “A model of heterogeneous undercooled liquid and glass accounting for temperature-dependent nonexponentiality and enthalpy fluctuation,” *J. Chem. Phys.* **160**, 174504 (2024). doi:10.1063/5.0196812
- W. Takeda and P. Lucas, “Relationship between enthalpy fluctuation and nonexponential relaxation in glass-forming liquids,” *Mater. Adv.* **7**, 5729 (2026). doi:10.1039/d6ma00442c

and, when you use them, the TNM and MAP / RelaxPy papers:

- A. Q. Tool, *J. Am. Ceram. Soc.* **29**, 240 (1946); O. S. Narayanaswamy, *J. Am. Ceram. Soc.* **54**, 491 (1971); C. T. Moynihan et al., *J. Am. Ceram. Soc.* **59**, 12 (1976); I. M. Hodge, *Macromolecules* **20**, 2897 (1987); G. W. Scherer, *J. Am. Ceram. Soc.* **69**, 374 (1986).
- J. C. Mauro, D. C. Allan and M. Potuzak, *Phys. Rev. B* **80**, 094204 (2009); X. Guo et al., *J. Am. Ceram. Soc.* **101**, 1169 (2018); J. C. Mauro and Y. Z. Mauro, *Physica A* **506**, 75 (2018); J. C. Mauro, R. J. Loucks and P. K. Gupta, *J. Am. Ceram. Soc.* **92**, 75 (2009); C. J. Wilkinson, Y. Z. Mauro and J. C. Mauro, *SoftwareX* **7**, 255 (2018).

Experiment references (Kovacs, Macedo–Napolitano, Hodge, Schawe, Hutchinson–Montserrat) are listed in Learn → Further reading.

Tf Studio · Lucas group, Department of Materials Science and Engineering, University of Arizona · [github.com/Wtakeda0129/tf-studio](https://github.com/Wtakeda0129/tf-studio)